

MIT 14.452 Economic Growth

Lecture 1

Questions & Evidence

A Beginner-Friendly Deep-Dive

Based on Daron Acemoglu's lecture notes · MIT OpenCourseWare · Fall 2016

Why are some countries rich and others poor?

This document explains MIT's Lecture 1 on Economic Growth in clear, accessible language for readers with a basic economics and math/engineering background. Every key idea is unpacked with analogies, context, and real-world examples.

Table of Contents

1.	What Is Economic Growth and Why Does It Matter?	3
2.	Cross-Country Income Differences — The Big Picture	3
3.	Reading the Data: GDP per Capita vs. GDP per Worker	4
4.	Growth Trajectories: Why Paths Diverge	4
5.	Proximate vs. Fundamental Causes	5
6.	Persistence and the Reversal of Fortune	5
7.	The Role of Institutions	6
8.	Does Democracy Cause Growth? The Econometrics	6
9.	Key Statistical Tools Used in This Lecture	8
1 0.	Summary & Big Takeaways	9
1 1.	Glossary of Key Terms	9
1 2.	References & Further Reading	10

SECTION 1 — What Is Economic Growth and Why Does It Matter?

1. What Is Economic Growth and Why Does It Matter?

Economic growth is the sustained increase in the amount of goods and services an economy produces over time — usually measured by Gross Domestic Product (GDP) per capita (the total output divided by the number of people). This lecture asks a deceptively simple question: **why are some countries much richer than others?**

Why Should You Care?

Higher income per capita is closely linked to better health outcomes, longer life expectancy, lower infant mortality, better education access, and higher individual well-being. A country that grows at 2% per year doubles its income every ~35 years; one that grows at 7% per year (like South Korea in the 1970s–90s) doubles income every ~10 years. Small differences in growth rates, sustained over decades, produce enormous gaps in living standards.

The 'Rule of 70' Analogy

A quick engineering-friendly fact: the number of years it takes for income to double equals roughly 70 divided by the annual growth rate. At 1% growth → 70 years to double. At 7% growth → 10 years to double. That is why sustained growth rates matter so much — the difference between India in 1980 and South Korea today is largely a story of different long-run growth rates maintained over decades.

SECTION 2 — Cross-Country Income Differences: The Big Picture

2. Cross-Country Income Differences — The Big Picture

The lecture opens with a striking world map showing GDP per capita in 2008. The colour gradient reveals a stark divide: North America, Western Europe, and Australia are wealthy (>20,000 per person); most of Sub-Saharan Africa and South Asia remain below 2,000. This isn't new — these gaps have been widening since the Industrial Revolution.

2.1 The Distribution of World Income

Acemoglu shows a chart of the **distribution of countries by GDP per capita** in 1960, 1980, and 2000. Two patterns stand out:

- **The average shifts right** — most countries grew richer over time.
- **The distribution spreads out** — the gap between rich and poor nations widened.

What is PPP adjustment?

Prices differ across countries — a haircut costs more in New York than in Lagos. Purchasing Power Parity (PPP) adjustment converts all incomes into a common currency that accounts for these price differences, allowing fair comparisons of real living standards. All income figures in this lecture are PPP-adjusted.

2.2 Why Use Log Income Instead of Income Levels?

Economists often plot the *logarithm* of income rather than income itself. Here's why:

When something grows **proportionally** (e.g., 5% per year), its logarithm grows **linearly**. If both a rich country and a poor country grow by 10%, their absolute dollar gap widens — but the *ratio* stays constant. On a log scale, equal vertical distances mean equal *percentage* differences, making it much easier to compare growth rates visually.

Log GDP

The natural logarithm of GDP. A one-unit increase = a ~170% rise in GDP. Equal vertical gaps on a log chart = equal percentage differences.

When Acemoglu replots the same distribution using **log GDP per capita**, the spreading-out is *less dramatic* — meaning part of what looked like growing inequality was actually just arithmetic from rising averages. On a log scale the picture is more nuanced: inequality rose, but more modestly than it first appeared.

SECTION 3 — GDP per Capita vs. GDP per Worker

3. Reading the Data: GDP per Capita vs. GDP per Worker

Acemoglu distinguishes between two related but different measures of economic performance:

Measure	What It Measures	Best Used For
GDP per capita	Total output ÷ total population	Comparing living standards
GDP per worker	Total output ÷ working population	Testing economic theory (productivity)

The reason economists prefer GDP per *worker* for theoretical analysis is that economic growth models predict how productive each worker is. Countries with many dependents (children, elderly) will have lower GDP per capita even if their workers are just as productive — so GDP per worker strips out demographic noise and gives a cleaner signal about the economy's productive capacity.

Looking at the log distribution of GDP per worker across 1960–2000, the lecture confirms the same two stylised facts: large inequality across nations, and a slight widening over time.

SECTION 4 — Growth Trajectories: Why Paths Diverge

4. Growth Trajectories: Why Paths Diverge

Acemoglu presents a chart of log GDP per capita from 1960–2000 for ten countries. The lines tell very different stories:

Country	Story	Key Question
USA	Steady growth from a high starting point	Why did the US stay ahead?
South Korea & Singapore	Rapid "miracle" growth — went from poor to rich in 40 years	How did they sustain ~7% growth for decades?
Spain	Fast growth ~1960–1985, then slowdown	Why did Spain converge then stall?
Brazil & Guatemala	Stagnation during the 1980s "lost decade"	What caused the decade-long slump?
Botswana	Surprisingly strong growth (diamonds + good institutions)	A bright spot in Africa — what was different?
Nigeria	Volatile and ultimately poor performance despite oil wealth	Why did oil fail to generate sustained growth?
India	Slow start, then acceleration post-1991 reforms	What unlocked India's growth potential?

The 200-year view (1820–2000) makes the divergence even starker. Before the Industrial Revolution, income differences across regions were modest. The Western Offshoots (USA, Canada, Australia, New Zealand) and Western Europe pulled dramatically ahead after ~1820, while Africa and parts of Asia lagged behind. This 'Great Divergence' is one of the central puzzles of economic history.

SECTION 5 — Proximate vs. Fundamental Causes

5. Proximate vs. Fundamental Causes of Growth

This is one of the most conceptually important parts of the lecture. Economists distinguish between two levels of explanation for why countries differ in income:

Proximate Causes	Fundamental Causes
Physical capital (machines, factories, infrastructure)	Geography (tropics, disease burden, coastal access)
Human capital (education, skills, health)	Culture (work ethic, trust, social norms)
Technology (productivity, innovation)	Institutions (property rights, rule of law, democracy)
"Why is South Korea richer?" → It has more capital, better-educated workers, and better technology.	"But why does South Korea have those things?" → That requires a fundamental explanation.

The Key Insight

Physical capital, human capital, and technology are *proximate* causes — they directly explain income differences. But they just push the question back one level: **why do some countries have more of these than others?** Answering *that* question requires understanding fundamental causes — especially institutions. The rest of the course is essentially about moving from proximate to fundamental explanations.

SECTION 6 — Persistence and the Reversal of Fortune

6. Persistence and the Reversal of Fortune

If geography were the main driver of prosperity, we'd expect countries with favourable geography (e.g., tropical fertile lands, large populations) to always be richer. But Acemoglu and co-authors show a fascinating **reversal of fortune**: the *richest* regions before colonization (around 1500 AD) are often among the *poorest* today.

6.1 How Do We Measure Prosperity Before GDP Data?

Modern national income accounts only go back to the mid-20th century. For earlier centuries, economists use two proxies:

- **Urbanisation rate**: Before industry, only places with agricultural surplus could feed large non-farming populations. High urbanisation $\approx$ more prosperous.
- **Population density**: Dense populations also signal agricultural productivity and economic organisation.

6.2 The Reversal

Plotting urbanisation in 1500 against GDP per capita in 1995, for former colonies, reveals a downward-sloping relationship — high urbanisation in 1500 predicts *lower* income today. This is the 'Reversal of Fortune.'

Examples: The Aztec/Inca empires (modern Mexico, Peru) were highly urbanised in 1500 but are middle-income today. The lands that became the USA and Canada were sparsely populated then but are among the wealthiest today.

6.3 Why Did This Reversal Happen?

The reversal coincides with the Industrial Revolution (~1750–1850). The timing chart in the lecture shows that countries with *low* urbanisation in 1500 industrialised rapidly after 1800, while previously-urban regions did not. The key mechanism was **colonialism**: in densely populated, wealthy regions, European colonisers set up extractive institutions (forced labour, resource extraction). In sparsely populated regions, they set up settler colonies with property rights and inclusive institutions — institutions that later enabled industrialisation.

Geography vs. Institutions

If geography (climate, disease, resources) were the main driver, the reversal of fortune is hard to explain — geography didn't change. But institutions *did* change dramatically with colonisation. This is strong evidence that institutions, not just geography, play a fundamental role in long-run development.

SECTION 7 — The Role of Institutions

7. The Role of Institutions

Institutions are the rules of the game in a society — formal laws and informal norms that structure political, economic, and social interactions. The lecture argues that institutions are the most important *fundamental cause* of economic growth. Three types of empirical evidence are presented:

Type	What It Does	Example
Cross-country long-run	Exploits exogenous variation to identify causal effects of institutions	Acemoglu, Johnson & Robinson (2001): settler mortality as instrument
Within-country	Compares localities within same country with different institutional histories	Dell (2010): the Peruvian mita (forced labour system)
Growth regressions	Statistical relationship between institutional quality & growth over shorter periods	Many studies using World Bank governance indicators

SECTION 8 — Does Democracy Cause Growth? The Econometrics

8. Does Democracy Cause Growth? The Econometrics

The most technically detailed part of the lecture concerns whether democracy — a key political institution — actually *causes* higher GDP. The conventional wisdom (that democracy is bad or irrelevant for growth) turns out to be wrong. Acemoglu et al. (2014) find that democratisation raises GDP per capita by roughly **20–30%** over the subsequent 20 years. Here's how they establish this rigorously.

8.1 The Core Challenge: Reverse Causality

The problem: countries don't democratise randomly. They tend to democratise when they are already in economic trouble — the chart in the lecture shows GDP falling in the years *before* democratisation. This means a naive comparison of democracies vs. autocracies will be biased: democracies might look like slow-growth countries simply because they democratised during bad economic times. Any honest analysis must account for this.

8.2 Method 1: Panel Data with Country Fixed Effects

The core regression model is:

$$y_{ct} = \beta \cdot D_{ct} + \sum \gamma_j y_{ct-j} + \alpha_c + \delta_t + \varepsilon_{ct}$$

Term	Meaning	Why It's Included
y_{ct}	Log GDP per capita in country c at time t	Outcome we are trying to explain
D_{ct}	1 if country c is a democracy in year t , 0 otherwise	The variable whose effect we want to measure
β	Coefficient on democracy	The causal effect we are trying to estimate
$\sum \gamma_j y_{ct-j}$	Lagged values of GDP (up to p years back)	Controls for the fact that democratisation happens when GDP is low
α_c	Country fixed effects (a dummy for each country)	Removes all time-invariant country characteristics (culture, geography)
δ_t	Year fixed effects (a dummy for each year)	Removes global shocks that affect all countries (oil crisis, recessions)
ε_{ct}	Error term	Residual variation unexplained by the model

Key result: Without lagged GDP controls (Column 1 of the results table), the coefficient on democracy is *negative* ($\beta = -10$). This is the biased result — it just reflects that countries democratise during downturns. Once lagged GDP is added (Columns 2–5), the coefficient flips positive and is statistically significant at roughly $\beta \approx 0.8$, implying a long-run GDP boost of roughly 20–22%.

8.3 Method 2: Semi-Parametric Matching

The panel regression assumes linear relationships. Semi-parametric matching relaxes this. It compares countries that democratised to a carefully constructed **matched control group** of non-democracies with similar GDP histories, using inverse propensity score weighting. The resulting 'event study' chart shows: GDP starts rising about 5 years after democratisation and continues upward for 25+ years. Conversely, reversals of democracy (coups) are associated with persistent GDP declines.

8.4 Method 3: Instrumental Variables (IV)

Even matching on observables might miss some unmeasured confounders. The IV approach finds a variable that *causes* democratisation but has no direct effect on GDP — this isolates the causal channel.

The instrument used is **regional democratisation waves**: when neighbouring countries in the same region (with similar political histories) democratise, a country is more likely to democratise too — a form of political contagion. This regional average (leaving out the country itself) predicts whether a country democratises but should not directly affect its GDP. Using this instrument still yields a large positive effect of democracy on long-run GDP.

The 'Persistence of GDP' Problem — Nickell Bias

Including lagged GDP on the right-hand side of a panel regression with fixed effects creates a well-known statistical problem called Nickell bias (after economist Simon Nickell). It biases coefficients when the panel has many countries but few time periods. Acemoglu et al. address this with alternative estimators (ABOND, HHK) and show the bias is small given their long time series — results are robust.

8.5 Mechanisms: How Does Democracy Raise GDP?

The data show that democracy raises GDP through several channels:

- **Higher investment share** in GDP (+9% long-run)
- **Economic reforms** (liberalisation, reduced distortions)
- **More trade openness** (+40% long-run)
- **Higher tax revenue** enabling public good provision (+16% long-run)
- **Better schooling** — primary and secondary enrolment rise
- **Lower child mortality** (−34% long-run)
- **Less political unrest** (−12% long-run)

SECTION 9 — Key Statistical Tools

9. Key Statistical Tools Used in This Lecture

This section gives a plain-English explanation of the statistical methods, for readers with a math/engineering background who are new to econometrics.

OLS Regression

Ordinary Least Squares: fits a straight line (or hyperplane) through data to minimise the sum of squared errors. The workhorse of econometrics. Problem: gives biased estimates when the explanatory variable (here: democracy) is correlated with unmeasured factors in the error term.

Panel Data / Fixed Effects

Data where we observe many units (countries) over many time periods. Fixed effects remove all time-invariant differences between units — think of it as subtracting each country's average from all its values, so we only use within-country variation over time.

Instrumental Variables (2SLS)

Two-stage least squares: first stage regresses the endogenous variable (democracy) on the instrument (regional democratisation); second stage uses the predicted values from the first stage in the main regression. This isolates only the exogenous variation in democracy driven by regional waves.

Semi-Parametric Matching

Instead of assuming a linear model, we compare treated units (democratising countries) to a weighted average of control units (non-democracies) matched on similar observable characteristics. Inverse propensity score weighting reweights controls to look like the treatment group in terms of pre-treatment GDP paths.

Nickell Bias

A bias that arises in dynamic panel models (with lagged dependent variable + fixed effects) when T (number of time periods) is small. The bias is of order $1/T$, so with $T \approx 50$ annual observations it is small. Addressed using GMM estimators (Arellano-Bond).

Long-Run Multiplier

In a dynamic model like this one, the short-run coefficient β understates the total effect. The long-run effect is $\beta / (1 - \Sigma \gamma_j)$, where $\Sigma \gamma_j$ measures the persistence of GDP. With persistence ≈ 0.963 , the long-run multiplier $\approx 1/(1-0.963) \approx 27\times$ the short-run β .

SECTION 10 — Summary & Big Takeaways

10. Summary & Big Takeaways

1	Income inequality across nations is enormous The richest countries are 30–50x wealthier per person than the poorest. This gap has widened since the Industrial Revolution.
2	Growth trajectories vary enormously Some countries (South Korea, Singapore) grew at 7%+ for 40 years. Others (Nigeria) stagnated despite natural resources. These differences compound dramatically.
3	Proximate causes are not the full story Capital, education, and technology explain growth mechanically but don't tell us why some societies accumulate these and others don't.
4	Institutions are fundamental Property rights, rule of law, political accountability — these determine whether individuals and firms can invest, innovate, and trade. The reversal of fortune is strong evidence that institutions, not geography, drive long-run prosperity.
5	Democracy boosts long-run growth by 20–30% Rigorous panel, matching, and IV evidence all point in the same direction. The mechanism runs through investment, reforms, education, and reduced mortality.
6	Controlling for dynamics matters Failing to account for GDP lags leads to spurious negative results. Proper econometrics — fixed effects, lagged controls, IV — reveals the true positive effect.

SECTION 11 — Glossary of Key Terms

11. Glossary of Key Terms

GDP per Capita	Total economic output divided by total population. Measures average living standard.
GDP per Worker	Total economic output divided by the employed/economically active population. Measures labour productivity.
PPP Adjustment	Purchasing Power Parity: adjusts income for price level differences so real living standards can be compared across countries.
Log Transformation	Taking the natural logarithm of income. Equal vertical distances on a log chart = equal percentage differences. Useful when growth is proportional.
Proximate Cause	A direct, mechanical explanation of an outcome. E.g., a country is rich because it has a lot of capital.
Fundamental Cause	A deeper explanation of why the proximate cause varies. E.g., a country has capital because its institutions protect property rights.
Institutions	The formal rules (laws, constitutions) and informal norms (culture, trust) that structure economic and political behaviour.
Reversal of Fortune	The empirical finding that previously prosperous regions (high urbanisation in 1500) are often poorer today — especially among former colonies.
Fixed Effects	Statistical controls that remove all time-invariant differences between units (e.g., countries). Equivalent to subtracting each country's mean from its data.
Instrumental Variable	A variable that affects the treatment (democracy) but has no direct effect on the outcome (GDP). Used to isolate causal variation.
Nickell Bias	A downward bias in dynamic panel models with lagged dependent variable and fixed effects, especially when T is small.
Long-Run Multiplier	The total cumulative effect of a change in a variable, after all dynamic adjustments have worked through the system.

Endogeneity	When an explanatory variable is correlated with the error term — usually due to reverse causality or omitted variables — leading to biased estimates.
2SLS	Two-Stage Least Squares: an IV estimator. Stage 1 regresses the endogenous variable on the instrument; Stage 2 uses the predicted values in the main equation.
Democratisation Wave	A period when multiple countries in a region transition to democracy near-simultaneously. Used here as an instrument for a single country's democratisation.

SECTION 12 — References & Further Reading

12. References & Further Reading

Primary Source

Acemoglu, D. (2014). *14.452 Economic Growth, Lecture 1: Questions and Evidence*. MIT OpenCourseWare.
<https://ocw.mit.edu/courses/14-452-economic-growth-fall-2016/>

Key Papers Cited in the Lecture

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- Acemoglu, D., Naidu, S., Restrepo, P., & Robinson, J.A. (2019). Democracy Does Cause Growth. *Journal of Political Economy*, 127(1), 47–100. [Working paper version cited as 2014 in lecture.]
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- Arellano, M., & Bond, S. (1991). Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations. *Review of Economic Studies*, 58(2), 277–297.

Textbooks & Further Reading

- Acemoglu, D. (2009). *Introduction to Modern Economic Growth*. Princeton University Press. [The companion textbook to this course — rigorous but comprehensive.]
- Acemoglu, D., & Robinson, J.A. (2012). *Why Nations Fail: The Origins of Power, Prosperity and Poverty*. Crown Publishers. [Accessible book-length treatment of the institutions thesis for a general audience.]
- Jones, C.I., & Vollrath, D. (2013). *Introduction to Economic Growth* (3rd ed.). W.W. Norton. [An excellent undergraduate-level textbook covering the same models.]
- Weil, D.N. (2012). *Economic Growth* (3rd ed.). Pearson. [Another accessible entry-level text with good empirics.]
- Maddison Project Database (2020). Long-run GDP data going back to 1820.
<https://www.rug.nl/ggdc/historicaldevelopment/maddison/> [The source of the long-run historical GDP charts in the lecture.]
- Penn World Tables (Version 10.0). Feenstra, R.C., Inklaar, R., & Timmer, M.P. (2015). The Next Generation of the Penn World Table. *American Economic Review*, 105(10), 3150–3182. [Primary source for PPP-adjusted cross-country GDP data.]

Online Resources

- Our World in Data – Economic Growth: <https://ourworldindata.org/economic-growth> [Interactive charts and accessible explanations of growth facts.]
- MIT OpenCourseWare 14.452: <https://ocw.mit.edu/courses/14-452-economic-growth-fall-2016/> [All lecture slides, problem sets, and readings freely available.]
- World Bank Open Data: <https://data.worldbank.org> [GDP per capita, PPP, and thousands of other development indicators.]

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